

MH2100 Calculus III

Midterm I (Solutions)

AY 2021/2022, Semester 1

Quantitative Research Society @NTU

September 2021

Overview

- **Coverage:** parametric curves and tangent lines; limits, continuity and partial derivatives; directional derivatives, tangent planes, the Chain Rule, implicit differentiation, and critical points.
- **Format:** three questions, 50 marks in total, 90 minutes (official paper: September 2021).
- **Historical permitted material:** the September 2021 paper permitted one two-sided A4 piece of paper with handwritten notes. Consult current course materials for present assessment rules.
- **Source:** MH2100 Midterm I, AY2021/2022 Semester 1. Questions are transcribed from the official paper; solutions and marking schemes are unofficial QRS@NTU material.

Instructions (verbatim)

1. DO NOT TURN OVER PAPER UNTIL INSTRUCTED.
2. This midterm paper contains THREE (3) questions.
3. Answer ALL questions. The marks for each question are indicated at the beginning of each question.
4. Candidates can write anywhere on this midterm paper.
5. ONE two-sided A4 piece of paper with handwritten notes is permitted.

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Question 1**(17 Marks)****QUESTION 1.**

- (i) [3 marks] Find parametric equations for the line that is tangent to the curve parametrised by

$$\mathbf{r}(t) = \begin{pmatrix} 5 \sin t \\ t^3 \\ 4e^{3t} \end{pmatrix},$$

at the point $(0, 0, 4)$.

- (ii) [3 marks] The cartesian equation $y = ax + b$ describes the line tangent to the curve

$$\begin{cases} x(t) = \sin(2t), \\ y(t) = t \cos t \end{cases}$$

at the point defined by $t = \pi/2$. Find the values of a and b .

- (iii) [3 marks] At the point defined by $t = a$ the tangent vector of the curve parametrised by

$$\mathbf{r}(t) = \begin{pmatrix} e^{\sin t} \tan t \\ t^2 - 10t \\ 23 \end{pmatrix},$$

is orthogonal to the vector $\langle 0, 1, 2 \rangle$. Find the value of a .

- (iv) [3 marks] Find the directional derivative of the function

$$f(x, y, z) = x^3 + y^2 - 4z^2 + z \ln x$$

in the direction of the vector $\langle 1, 4, 1 \rangle$.

- (v) [5 marks] In which directions is the directional derivative of

$$f(x, y) = x^2y - y^8 + 2x$$

at $(6, 1)$ equal to zero? Write the unit vector for each direction.

Solution**Method 1: Direct computation**

- (i) The point condition forces $t = 0$, since $t^3 = 0$. Differentiate:

$$\mathbf{r}'(t) = \begin{pmatrix} 5 \cos t \\ 3t^2 \\ 12e^{3t} \end{pmatrix}, \quad \mathbf{r}'(0) = \langle 5, 0, 12 \rangle.$$

Therefore the tangent line through $(0, 0, 4)$ is

$$\boxed{\mathbf{l}(s) = \langle 0, 0, 4 \rangle + s \langle 5, 0, 12 \rangle, \quad s \in \mathbb{R},}$$

or, equivalently, $\boxed{x = 5s, \ y = 0, \ z = 4 + 12s.}$

(ii) At $t = \pi/2$,

$$x = \sin \pi = 0, \quad y = \frac{\pi}{2} \cos \frac{\pi}{2} = 0.$$

Moreover,

$$\frac{dx}{dt} = 2 \cos(2t), \quad \frac{dy}{dt} = \cos t - t \sin t,$$

so at $t = \pi/2$,

$$\frac{dy}{dx} = \frac{-\pi/2}{-2} = \frac{\pi}{4}.$$

The tangent line passes through the origin. Hence

$$a = \frac{\pi}{4}, \quad b = 0.$$

(iii) Orthogonality to $\langle 0, 1, 2 \rangle$ only uses the second and third components of the tangent vector:

$$\mathbf{r}'(t) \cdot \langle 0, 1, 2 \rangle = (2t - 10) + 2(0) = 2t - 10.$$

Thus $2a - 10 = 0$, and

$$a = 5.$$

(iv) The unit vector in the stated direction is

$$\mathbf{u} = \frac{\langle 1, 4, 1 \rangle}{\sqrt{18}}.$$

Since

$$\nabla f(x, y, z) = \left\langle 3x^2 + \frac{z}{x}, 2y, -8z + \ln x \right\rangle,$$

the directional derivative is

$$D_{\mathbf{u}}f(x, y, z) = \frac{3x^2 + z/x + 8y - 8z + \ln x}{\sqrt{18}} \quad (x > 0).$$

(v) At $(6, 1)$,

$$\nabla f(x, y) = \langle 2xy + 2, x^2 - 8y^7 \rangle, \quad \nabla f(6, 1) = \langle 14, 28 \rangle = 14\langle 1, 2 \rangle.$$

For a unit direction $\mathbf{u} = \langle u_1, u_2 \rangle$, the condition $D_{\mathbf{u}}f(6, 1) = 0$ is

$$u_1 + 2u_2 = 0, \quad u_1^2 + u_2^2 = 1.$$

The two unit vectors orthogonal to $\langle 1, 2 \rangle$ are

$$\mathbf{u} = \frac{1}{\sqrt{5}}\langle 2, -1 \rangle \quad \text{or} \quad \mathbf{u} = \frac{1}{\sqrt{5}}\langle -2, 1 \rangle.$$

Unofficial QRS marking scheme (17 marks)

- (i) **(3 marks)** 1: identifies $t = 0$; 1: obtains $\mathbf{r}'(0) = \langle 5, 0, 12 \rangle$; 1: gives the line through $(0, 0, 4)$.
- (ii) **(3 marks)** 1: finds the point $(0, 0)$; 1: computes slope $\pi/4$; 1: gives $a = \pi/4$, $b = 0$.
- (iii) **(3 marks)** 1: differentiates the relevant components; 1: imposes orthogonality; 1: obtains $a = 5$.
- (iv) **(3 marks)** 1: computes the gradient; 1: normalizes the direction; 1: takes the correct dot product.
- (v) **(5 marks)** 2: computes $\nabla f(6, 1) = \langle 14, 28 \rangle$; 1: states orthogonality; 1: obtains one correct unit direction; 1: obtains the opposite direction.

Question 2**(16 Marks)****QUESTION 2.**

- (i) [8 marks] Evaluate each of the following limits or show that the limit does not exist. Justify your answers.

(a)

$$\lim_{(x,y) \rightarrow (5,0)} \frac{5x^2y + 10}{\sqrt{x^2 + y^2}}$$

(b)

$$\lim_{(x,y) \rightarrow (0,0)} \frac{3y}{\sqrt{x^2 + y^2}}$$

(c)

$$\lim_{(x,y) \rightarrow (0,0)} \frac{3x^2 + x^2y + xy^3 + 3y^2}{x^2 + y^2}$$

- (ii) [8 marks] Define

$$f(x, y) = \begin{cases} \frac{x^3y + y^3}{x^6 + y^2}; & \text{if } x^2 + y^2 \neq 0, \\ 0; & \text{if } x^2 + y^2 = 0. \end{cases}$$

- (a) Is f continuous at $(0, 0)$? Justify your answer.
 (b) Find $f_x(0, 0)$ and $f_y(0, 0)$.
 (c) Is f differentiable at $(0, 0)$? Justify your answer.

Solution**Method 1: Direct limits and definitions**

- (i) (a) The denominator tends to $\sqrt{25} = 5 \neq 0$, so the quotient is continuous at $(5, 0)$. Direct substitution gives

$$\lim_{(x,y) \rightarrow (5,0)} \frac{5x^2y + 10}{\sqrt{x^2 + y^2}} = \frac{10}{5} = 2.$$

- (b) Along $y = 0$ with $x \neq 0$, the quotient is 0. Along $x = 0$ with $y \neq 0$, it is

$$\frac{3y}{\sqrt{y^2}} = \frac{3y}{|y|},$$

which tends to 3 as $y \rightarrow 0^+$ and to -3 as $y \rightarrow 0^-$. Therefore

the limit does not exist.

- (c) Write the quotient as

$$3 + \frac{x^2y}{x^2 + y^2} + \frac{xy^3}{x^2 + y^2}.$$

With $r = \sqrt{x^2 + y^2}$,

$$\left| \frac{x^2y}{x^2 + y^2} \right| \leq |y| \rightarrow 0, \quad \left| \frac{xy^3}{x^2 + y^2} \right| \leq \frac{r^4}{r^2} = r^2 \rightarrow 0.$$

Hence

$$\lim_{(x,y) \rightarrow (0,0)} \frac{3x^2 + x^2y + xy^3 + 3y^2}{x^2 + y^2} = 3.$$

(ii) (a) Along $x = 0$,

$$f(0, y) = \frac{y^3}{y^2} = y \longrightarrow 0.$$

Along $y = x^3$ with $x \neq 0$,

$$f(x, x^3) = \frac{x^6 + x^9}{x^6 + x^6} = \frac{1 + x^3}{2} \longrightarrow \frac{1}{2}.$$

The path limits differ, so

$$f \text{ is not continuous at } (0, 0).$$

(b) By the definitions of the partial derivatives,

$$f_x(0, 0) = \lim_{h \rightarrow 0} \frac{f(h, 0) - 0}{h} = 0,$$

and, since $f(0, k) = k$ for $k \neq 0$,

$$f_y(0, 0) = \lim_{k \rightarrow 0} \frac{f(0, k) - 0}{k} = 1.$$

Thus

$$f_x(0, 0) = 0, \quad f_y(0, 0) = 1.$$

(c) Differentiability at a point implies continuity there. Part (a) proves that f is not continuous at $(0, 0)$; hence

$$f \text{ is not differentiable at } (0, 0).$$

Unofficial QRS marking scheme (16 marks)

(i) (8 marks)

- (a) (2 marks) 1: continuity/substitution justification; 1: final value 2.
- (b) (3 marks) 1: computes the $y = 0$ path; 1: uses $x = 0$ to obtain distinct one-sided values; 1: concludes nonexistence.
- (c) (3 marks) 1: separates the constant term; 1: bounds both remaining terms; 1: concludes 3.

(ii) (8 marks)

- (a) (3 marks) 1: path $x = 0$; 1: path $y = x^3$; 1: concludes non-continuity.
- (b) (3 marks) 1: correct difference quotient for f_x ; 1: correct difference quotient for f_y ; 1: states both values.
- (c) (2 marks)
 - 1 mark: invokes differentiability implies continuity.
 - 1 mark: concludes non-differentiability.

Question 3**(17 Marks)****QUESTION 3.**

- (i) [3 marks] Find the directional derivative of the function

$$f(x, y) = \begin{cases} \frac{xy + y^2}{x^2 + y^2}; & \text{if } x^2 + y^2 \neq 0, \\ 0; & \text{if } x^2 + y^2 = 0, \end{cases}$$

in the direction of the vector $\langle 3, 4 \rangle$ at the point $(0, 0)$.

- (ii) [3 marks] Find an equation for the tangent plane to the surface given by the equation
- $z = 5x^2 - 10y^2$
- at the point
- $(2, 1, 10)$
- .

- (iii) [3 marks] Use the Chain Rule to find
- $\frac{\partial z}{\partial u}$
- when
- $u = 0$
- and
- $v = 2$
- for

$$z = \sin(xy) + \sin(y), \quad x = 2u^2 + 3v^2, \quad y = 2uv.$$

- (iv) [3 marks] Find
- $\frac{\partial z}{\partial y}$
- at the point
- $(\pi, 2\pi, \pi/2)$
- where
- z
- is defined implicitly by

$$3 \cos(x + 2y) + 3 \cos(x + 2z) + \cos(2y + z) = 0.$$

- (v) [5 marks] Let

$$f(x, y) = 27x^3 - 81x + 2y^2 + 8y - 17.$$

Find and classify all critical points of f . (For each critical point, determine whether or not it is a local max/min or a saddle point.)**Solution****Method 1: Direct computation**

- (i) Normalize the stated direction:

$$\mathbf{u} = \left\langle \frac{3}{5}, \frac{4}{5} \right\rangle.$$

For $t \neq 0$,

$$f(t\mathbf{u}) = f\left(\frac{3t}{5}, \frac{4t}{5}\right) = \frac{(3t/5)(4t/5) + (4t/5)^2}{(3t/5)^2 + (4t/5)^2} = \frac{28}{25}.$$

Consequently the directional-difference quotient is

$$\frac{f(t\mathbf{u}) - f(0, 0)}{t} = \frac{28}{25t},$$

which has no finite two-sided limit as $t \rightarrow 0$. Therefore

$$\boxed{D_{\mathbf{u}}f(0, 0) \text{ does not exist.}}$$

- (ii) Let
- $g(x, y) = 5x^2 - 10y^2$
- . Then

$$g_x = 10x, \quad g_y = -20y,$$

so $g_x(2, 1) = 20$ and $g_y(2, 1) = -20$. The tangent plane is

$$z - 10 = 20(x - 2) - 20(y - 1).$$

Thus an equation of the plane is

$$\boxed{20x - 20y - z = 10.}$$

(iii) The Chain Rule gives

$$\frac{\partial z}{\partial u} = z_x \frac{\partial x}{\partial u} + z_y \frac{\partial y}{\partial u},$$

where

$$z_x = y \cos(xy), \quad z_y = x \cos(xy) + \cos y,$$

and

$$x_u = 4u, \quad y_u = 2v.$$

At $(u, v) = (0, 2)$, we have $(x, y) = (12, 0)$, so

$$z_x = 0, \quad z_y = 12 + 1 = 13, \quad x_u = 0, \quad y_u = 4.$$

Hence

$$\left. \frac{\partial z}{\partial u} \right|_{(u,v)=(0,2)} = 0 \cdot 0 + 13 \cdot 4 = 52.$$

(iv) Put

$$F(x, y, z) = 3 \cos(x + 2y) + 3 \cos(x + 2z) + \cos(2y + z).$$

Implicit differentiation gives $F_y + F_z z_y = 0$, with

$$F_y = -6 \sin(x + 2y) - 2 \sin(2y + z),$$

$$F_z = -6 \sin(x + 2z) - \sin(2y + z).$$

At $(x, y, z) = (\pi, 2\pi, \pi/2)$,

$$\sin(x + 2y) = \sin(5\pi) = 0, \quad \sin(x + 2z) = \sin(2\pi) = 0, \quad \sin(2y + z) = \sin \frac{9\pi}{2} = 1.$$

Therefore $F_y = -2$, $F_z = -1$, and

$$\frac{\partial z}{\partial y} = -\frac{F_y}{F_z} = -2.$$

(v) The first partial derivatives are

$$f_x = 81x^2 - 81 = 81(x^2 - 1), \quad f_y = 4y + 8.$$

Thus $f_x = f_y = 0$ gives the two critical points

$$(1, -2) \quad \text{and} \quad (-1, -2).$$

The second partial derivatives are

$$f_{xx} = 162x, \quad f_{xy} = 0, \quad f_{yy} = 4,$$

so the discriminant is

$$D = f_{xx}f_{yy} - (f_{xy})^2 = 648x.$$

At $(1, -2)$, $D = 648 > 0$ and $f_{xx} = 162 > 0$, so it is a local minimum. At $(-1, -2)$, $D = -648 < 0$, so it is a saddle point. Hence

$$(1, -2) \text{ is a local minimum; } (-1, -2) \text{ is a saddle point.}$$

Unofficial QRS marking scheme (17 marks)

- (i) **(3 marks)** 1: normalizes the direction; 1: evaluates the function along that ray; 1: concludes that the directional derivative does not exist.
- (ii) **(3 marks)** 1: finds both partial derivatives; 1: evaluates the slopes at $(2, 1)$; 1: gives the correct plane.
- (iii) **(3 marks)** 1: writes the Chain Rule; 1: evaluates the intermediate variables and derivatives; 1: obtains 52.
- (iv) **(3 marks)** 1: forms F_y and F_z ; 1: evaluates them at the stated point; 1: obtains $z_y = -2$.
- (v) **(5 marks)**
- 2 marks: computes and solves $f_x = f_y = 0$.
 - 1 mark: computes the Hessian entries and discriminant.
 - 1 mark: classifies $(1, -2)$.
 - 1 mark: classifies $(-1, -2)$.